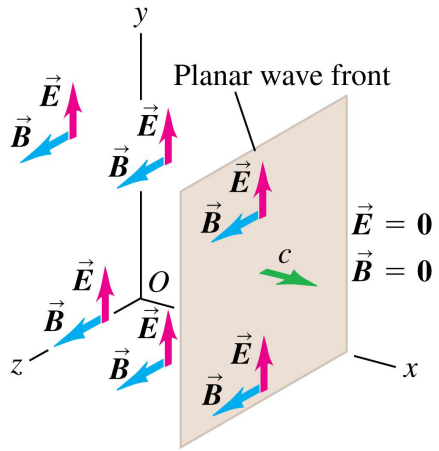
# In the previous lectures

- Maxwell's eq. & EM waves

$$E = cB$$

$$B = \epsilon_0 \mu_0 cE$$

$$c = \frac{1}{\sqrt{\epsilon_0 \mu_0}}$$

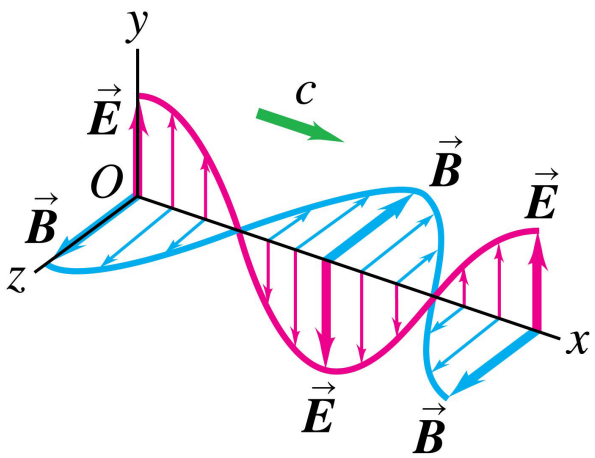



- Sinusoidal EM waves

$$\vec{E}(x,t) = \hat{j} E_{\max} \cos(kx - \omega t)$$

$$\vec{B}(x,t) = \hat{k} B_{\max} \cos(kx - \omega t)$$

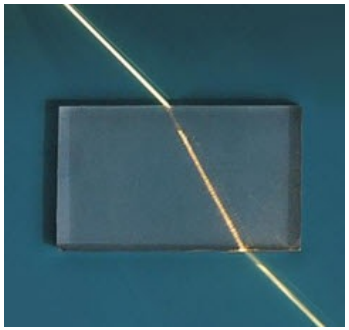
$$E_{\max} = cB_{\max}$$




- EM waves in matter

$$v = \frac{1}{\sqrt{\epsilon \mu}} = \frac{1}{\sqrt{K K_m}} \frac{1}{\sqrt{\epsilon_0 \mu_0}}$$

$$= \frac{c}{\sqrt{K K_m}}$$



© 2016 Pearso

# In the previous lecturesary

- Energy and momentum in EM waves

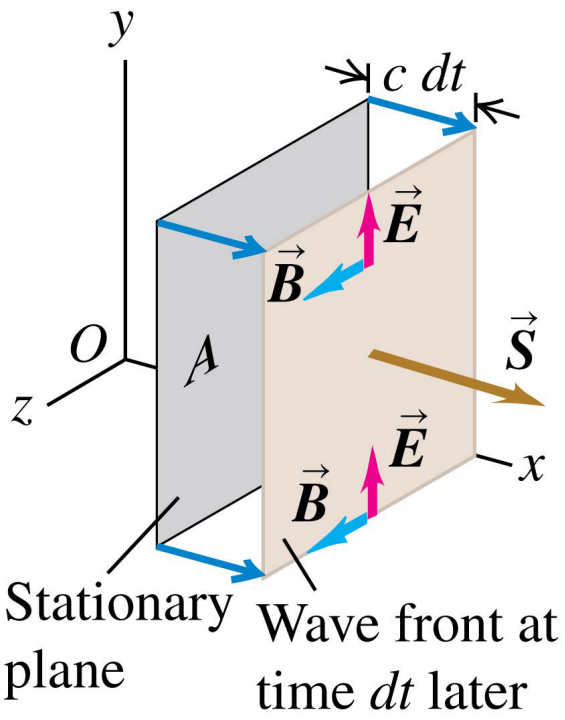
$$\vec{S} = \frac{1}{\mu_0}\vec{E} \times \vec{B}$$

$$I = S_{\text{av}} = \frac{E_{\text{max}}B_{\text{max}}}{2\mu_0} = \frac{E_{\text{max}}^2}{2\mu_0 c}$$

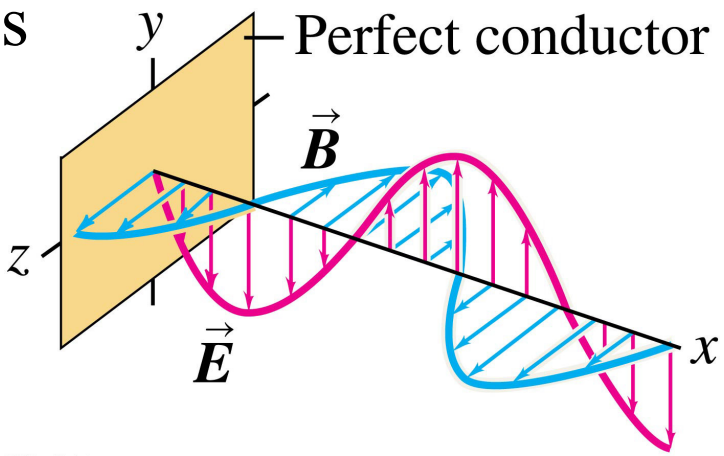
$$= \tfrac{1}{2}\sqrt{\frac{\epsilon_0}{\mu_0}}E_{\text{max}}^2$$

$$= \tfrac{1}{2}\epsilon_0 c E_{\text{max}}^2$$

$$\frac{1}{A}\frac{dp}{dt} = \frac{S}{c} = \frac{EB}{\mu_0 c}$$




- Standing EM waves




© 2016 Pearson Education, Ltd.

# Ch33 The Nature and Propagation of Light



- When a cut diamond is illuminated with white light, it sparkles brilliantly with a spectrum of vivid colors.
- These distinctive visual features are a result of light traveling much slower in diamond than in air, and that light of different colors travels at different speeds in diamond.
- But by studying the branch of physics called **optics**, we can understand the blue color of the sky and the design of optical devices such as telescopes, microscopes, cameras, eyeglasses, and the human eye.

© 2016 Pearson Education, Ltd.

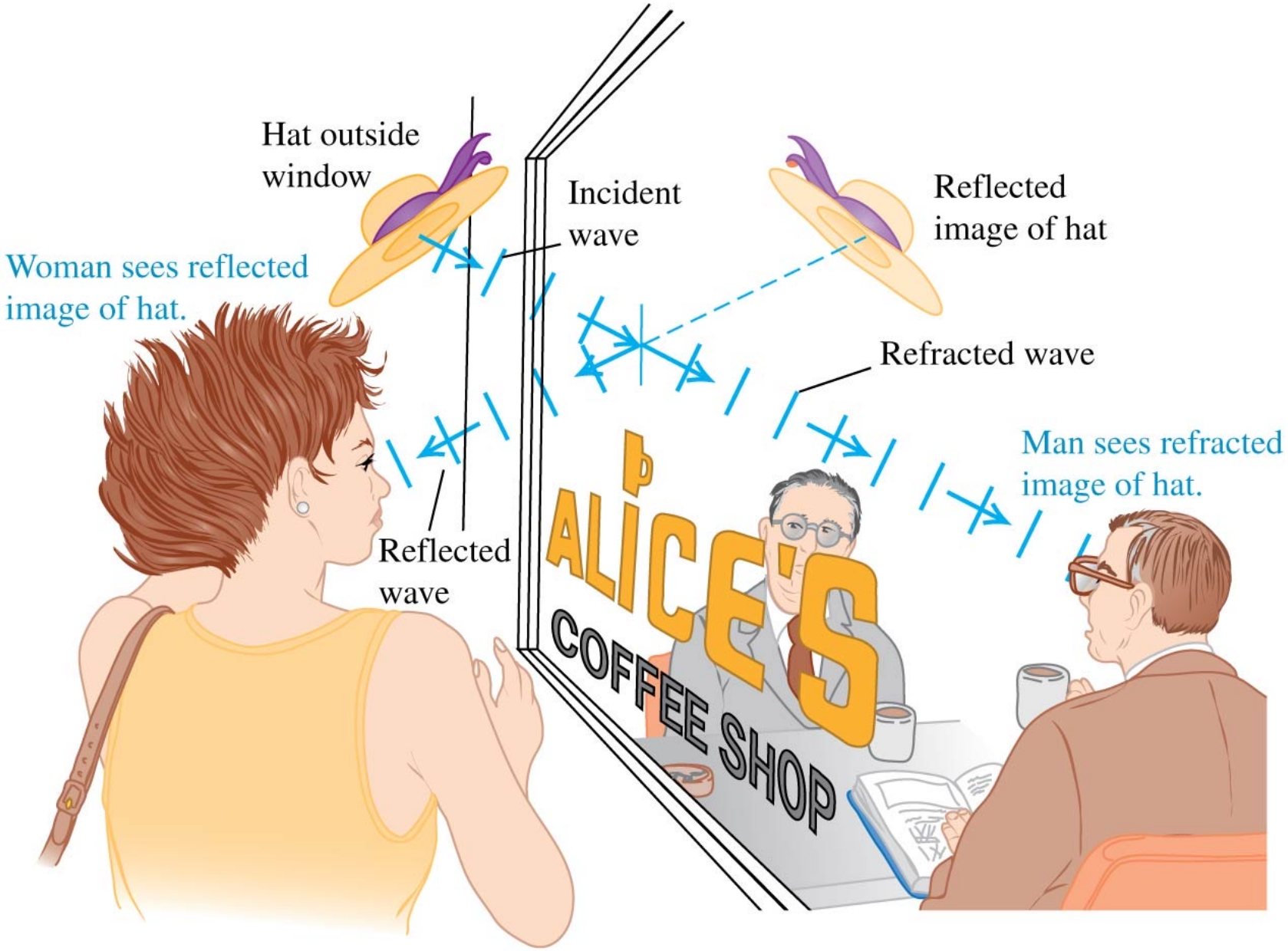
# Learning Goals for Chapter 33

***Looking forward at …***

- what light rays are, and how they are related to wave fronts.
- the laws that govern the reflection and refraction of light.
- the circumstances under which light is totally reflected at an interface.
- how to make polarized light out of ordinary light.
- how the scattering of light explains the blue color of the sky.
- how Huygens's principle helps us analyze reflection and refraction.

© 2016 Pearson Education, Ltd.

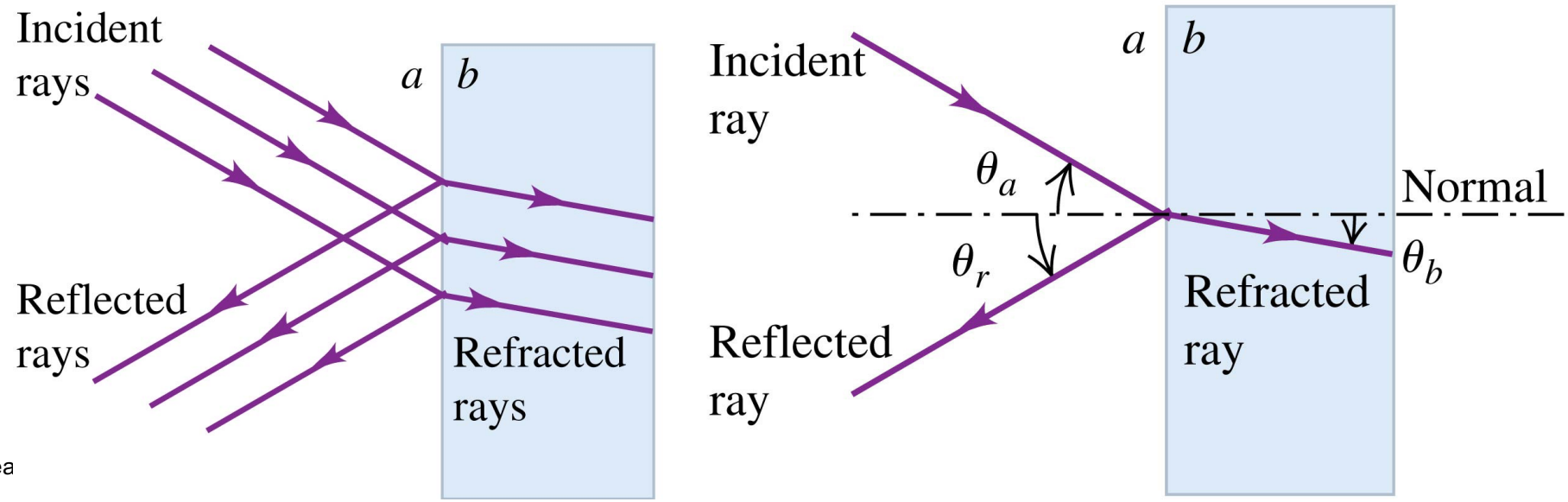
# Reflection and refraction




© 2016 Pearson Education, Ltd.

# Reflection and refraction

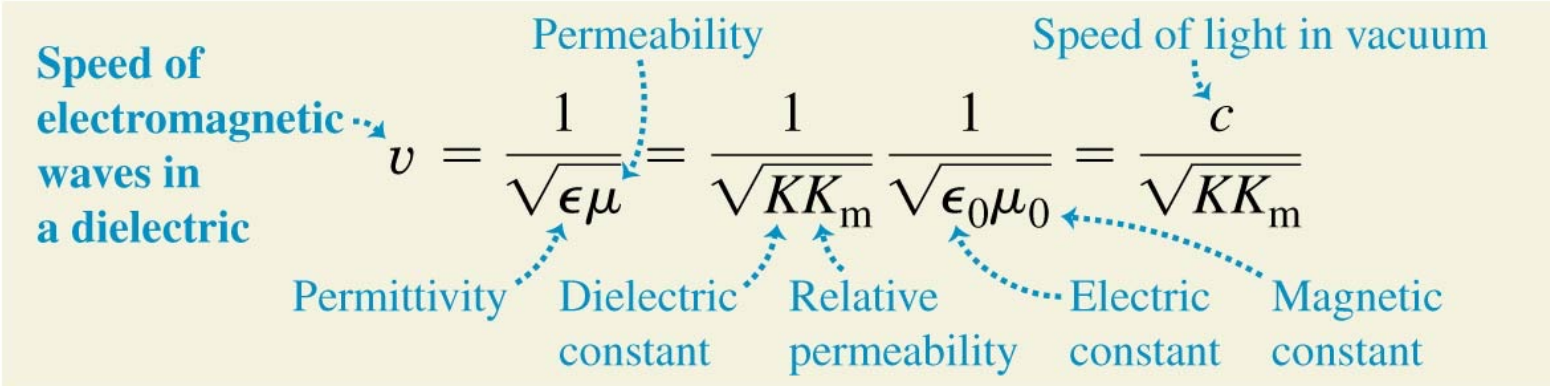
- When a light wave strikes a smooth interface separating two transparent materials (such as air and glass or water and glass), the wave is in general partly reflected and partly refracted (transmitted) into the second material.
- The segments of plane waves can be represented by bundles of rays forming beams of light.
- For simplicity we often draw only one ray in each beam.




© 2016 Pea

# Electromagnetic waves in matter

- Electromagnetic waves can travel in certain types of matter, such as air, water, or glass.
- When electromagnetic waves travel in nonconducting materials—that is, dielectrics—the speed $v$ of the waves depends on the dielectric constant of the material.



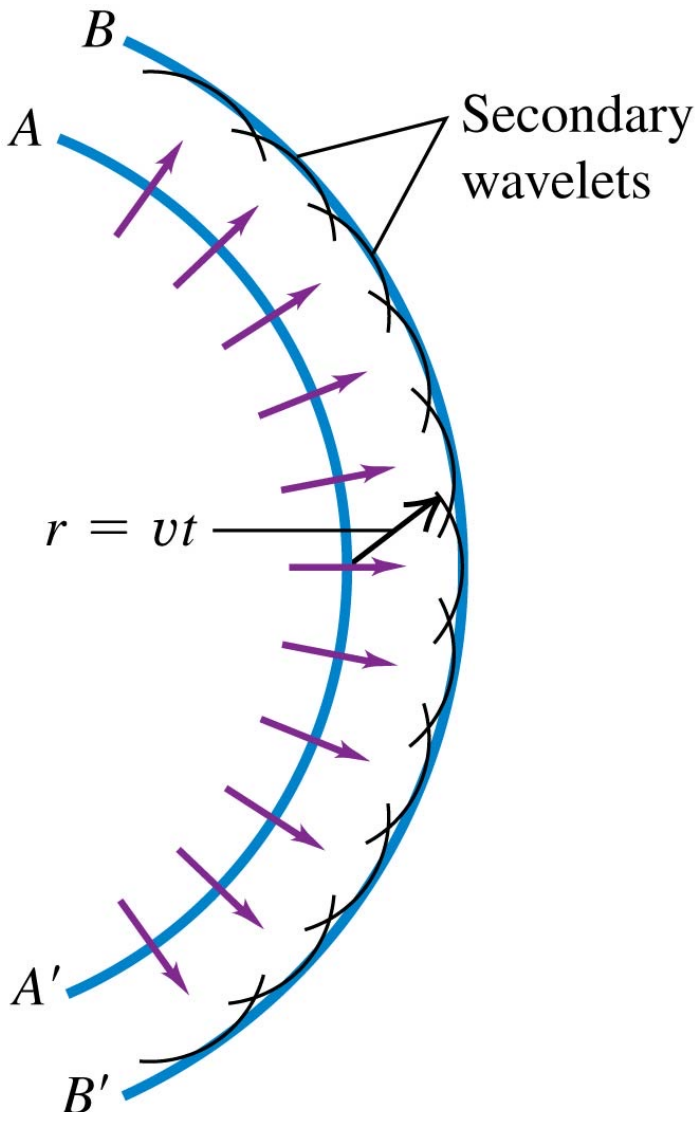

- The ratio of the speed $c$ in vacuum to the speed $v$ in a material is known in optics as the **index of refraction** $n$ of the material.

$$\frac{c}{v} = n = \sqrt{KK_m} \cong \sqrt{K}$$

© 2016 Pearson Education, Ltd.

# Huygens's principle

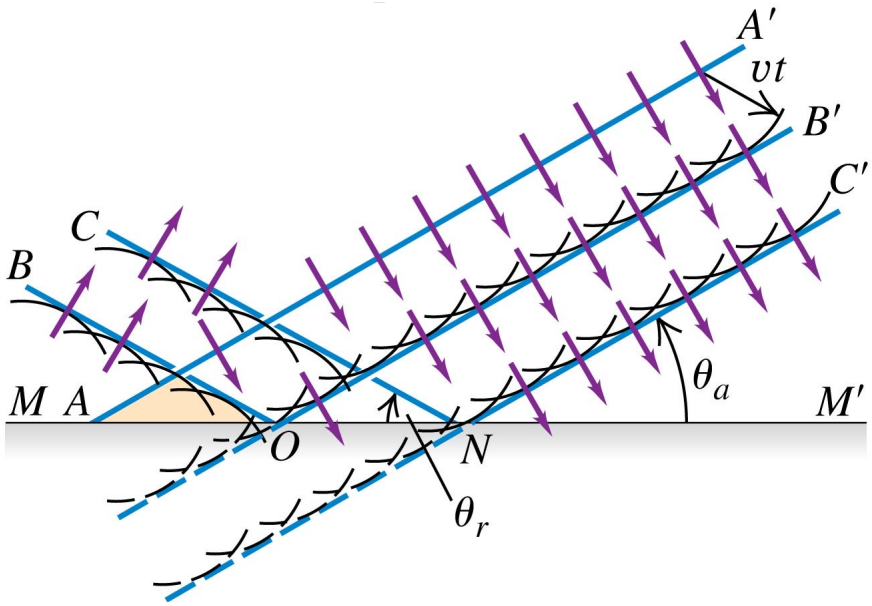
- **Huygens's principle** states that **every point of a wave front may be considered the source of secondary wavelets that spread out in all directions with a speed equal to the speed of propagation of the wave**.
- The new wave front at a later time is then found by constructing a surface *tangent* to the secondary wavelets or, as it is called, the *envelope* of the wavelets.
- The figure shows the application of Huygens's principle to wave front $AA'$ to construct a new wave front $BB'$.

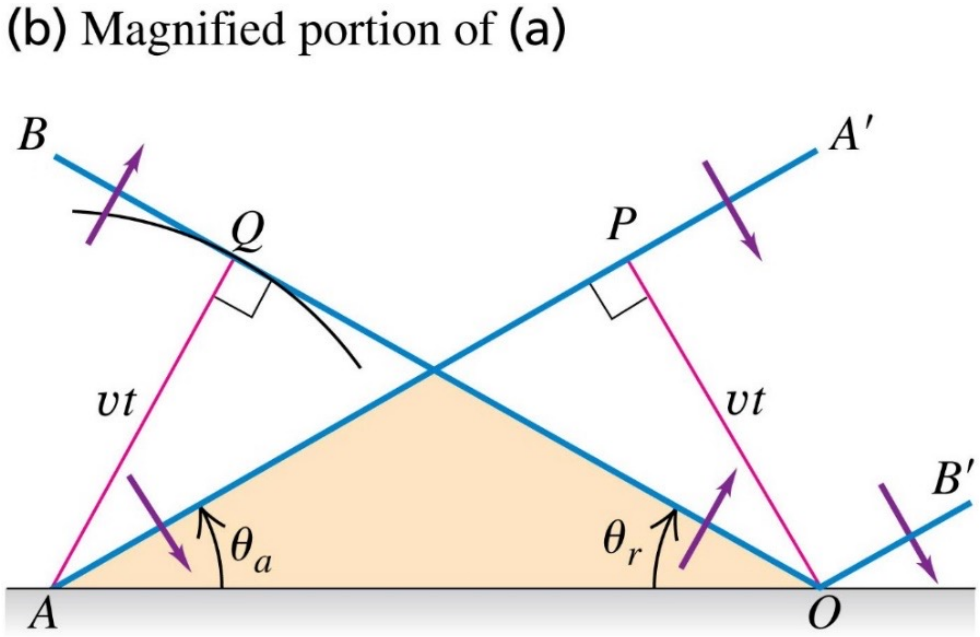



© 2016 Pearson Education, Ltd.

# Reflection and Huygens's principle

- To derive the law of reflection from Huygens's principle, we consider a plane wave approaching a plane reflecting surface.
- The effect of the reflecting surface is to change the direction of travel of those wavelets that strike it.
- The angle $\theta_a$ therefore equals the angle $\theta_r$, and we have the law of reflection.

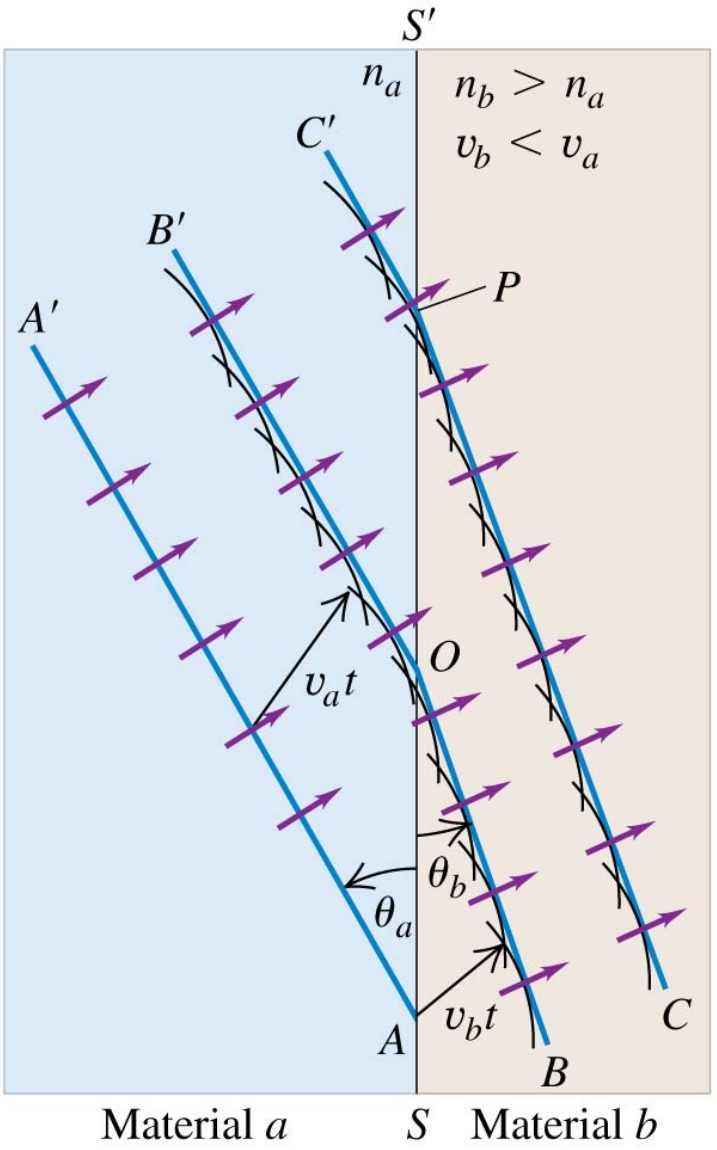


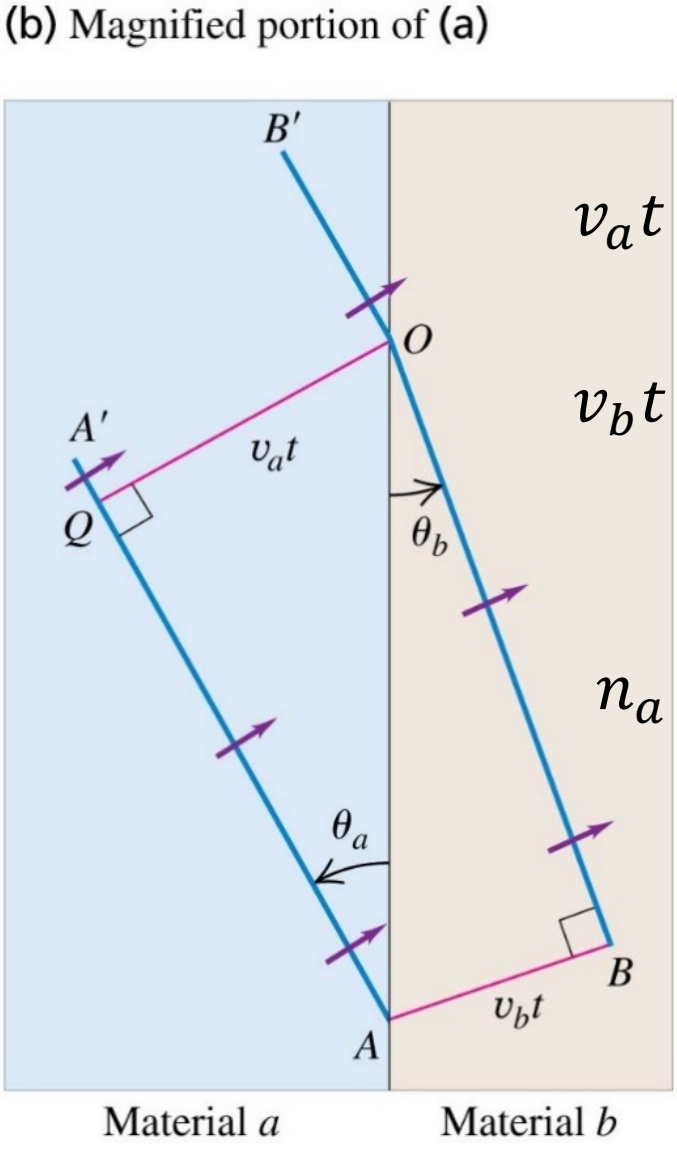




© 2016 Pearson Education, Ltd.

# Refraction and Huygens's principle

- Consider wave fronts traveling across the boundary surface $SS'$ between two transparent materials ***a*** and ***b*** ( $v_b < v_a$ ).




$$v_a t = \frac{c}{n_a} t = \overline{OA} \sin\theta_a$$

$$v_b t = \frac{c}{n_b} t = \overline{OA} \sin\theta_b$$

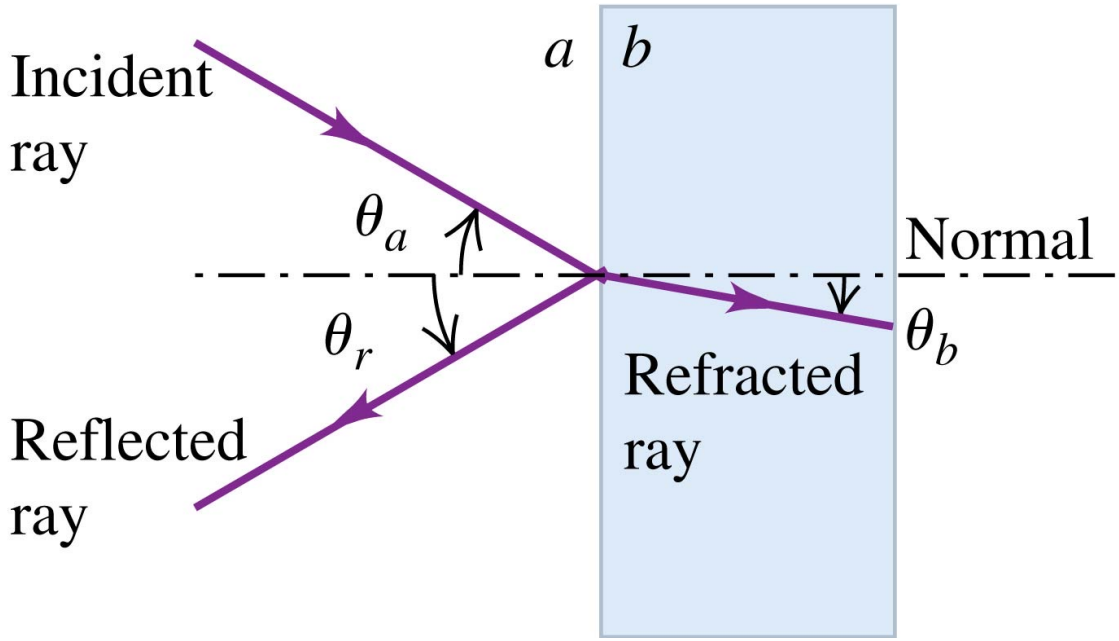
$$n_a \sin\theta_a = n_b \sin\theta_b$$

# The law of reflection

- The angle of reflection is equal to the angle of incidence for all wavelengths and for any pair of materials.

**Law of reflection:** $\theta_r = \theta_a$ (Angle of reflection (measured from normal) = Angle of incidence (measured from normal))

- Note that all angles are measured *from the normal*.




© 2016 Pearson Education, Ltd.

# Index of refraction

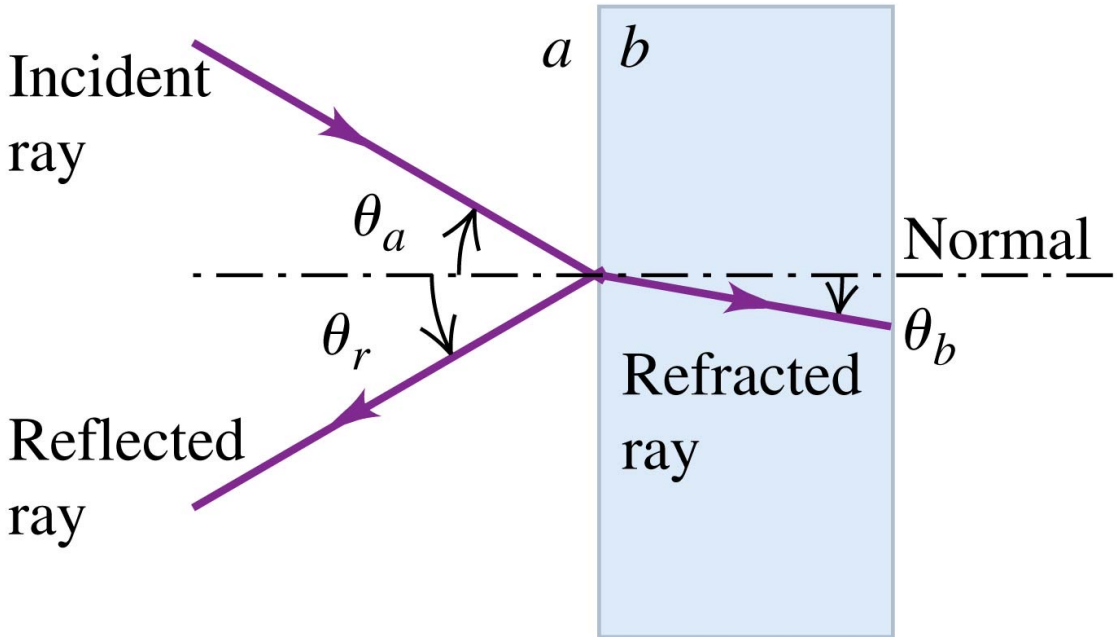
- The index of refraction of an optical material (also called the refractive index), denoted by $n$, is defined as:

**Index of refraction of an optical material** $n = \frac{c}{v}$

$c$: Speed of light in vacuum

$v$: Speed of light in the material

- For the case shown here, material ***b*** has a *larger* index of refraction than material ***a*** ($n_b > n_a$) and the angle $\theta_b$ is *smaller* than $\theta_a$.




© 2016 Pearson Education, Ltd.

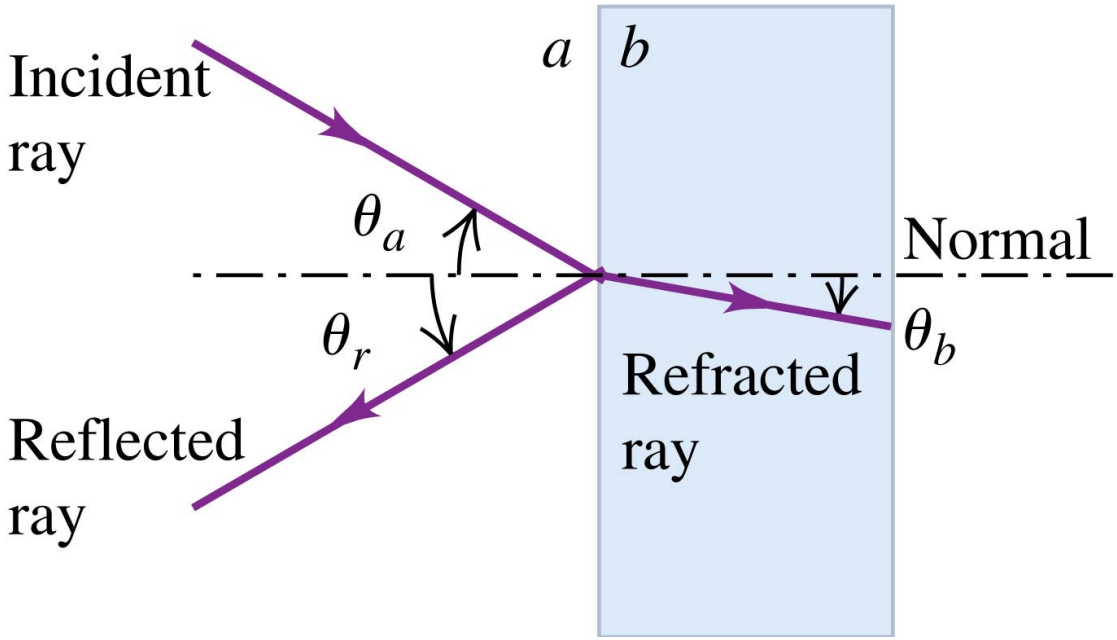
# The law of refraction

**Law of refraction:**

$$n_a \sin\theta_a = n_b \sin\theta_b$$

- $\theta_a$: Angle of incidence (measured from normal)
- $\theta_b$: Angle of refraction (measured from normal)
- $n_a$: Index of refraction for material with incident light
- $n_b$: Index of refraction for material with refracted light

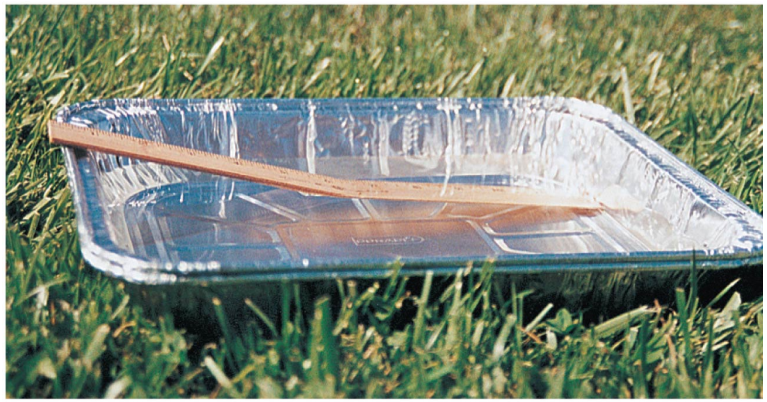
- This result is also called Snell's law, after the Dutch scientist Willebrord Snell (1591–1626).

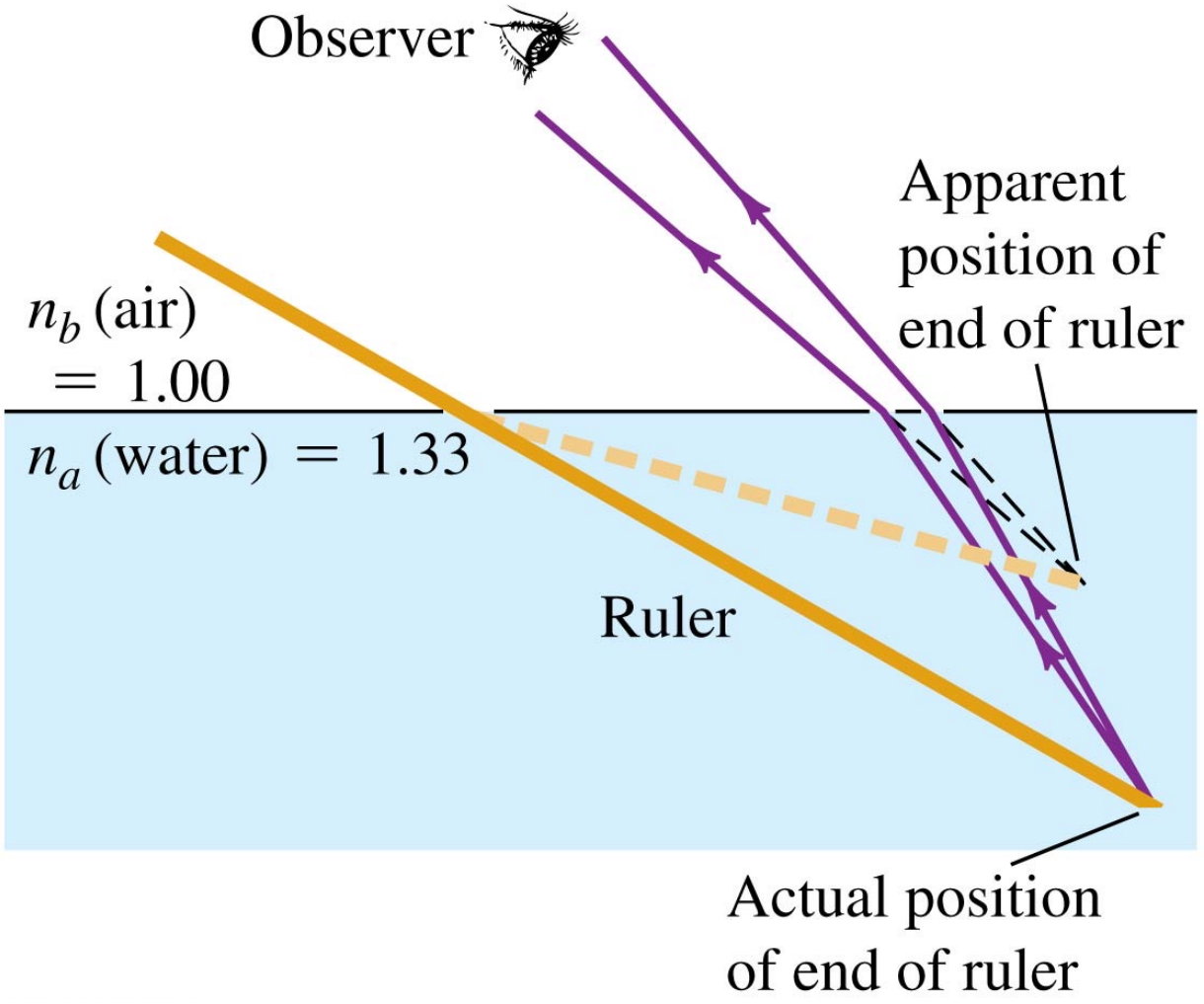



© 2016 Pearson Education, Ltd.

# Why does the ruler appear to be bent?

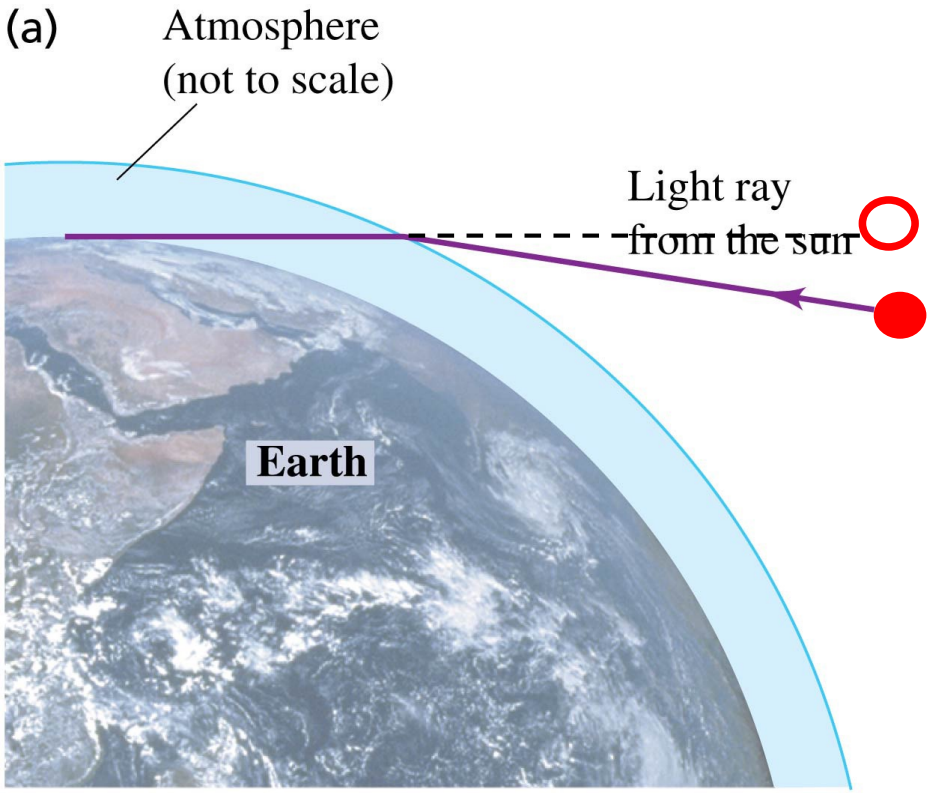
- Light rays coming from below the surface change in direction at the air–water interface, so the rays appear to be coming from a position above their actual point of origin.

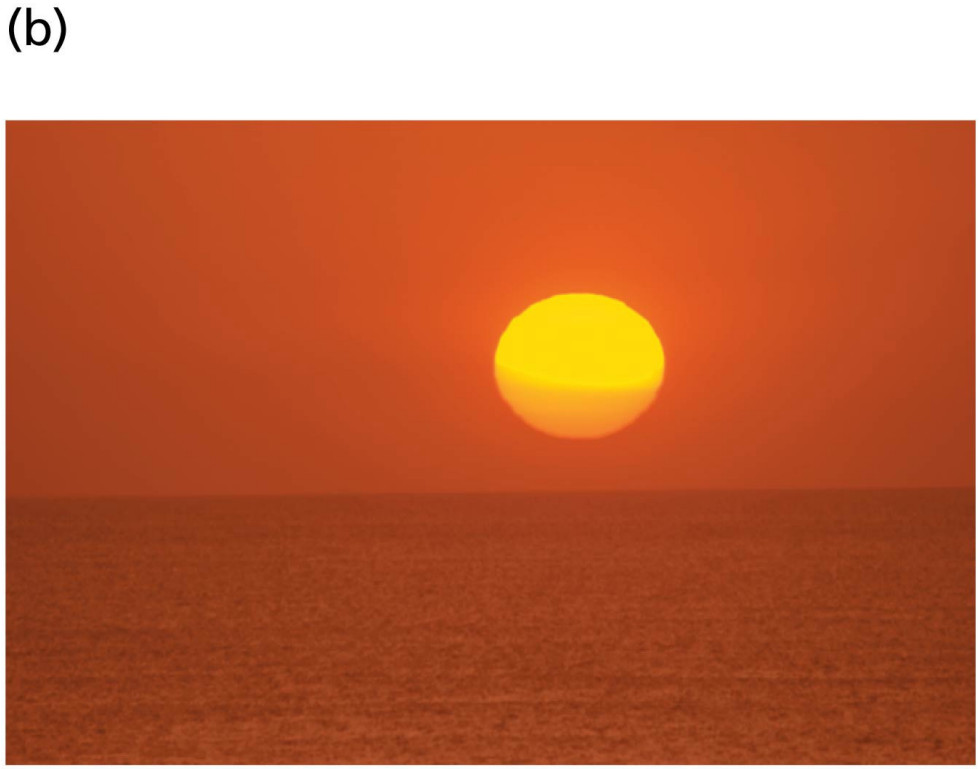





© 2016 Pearson Education, Ltd.

# The appearance of the setting sun



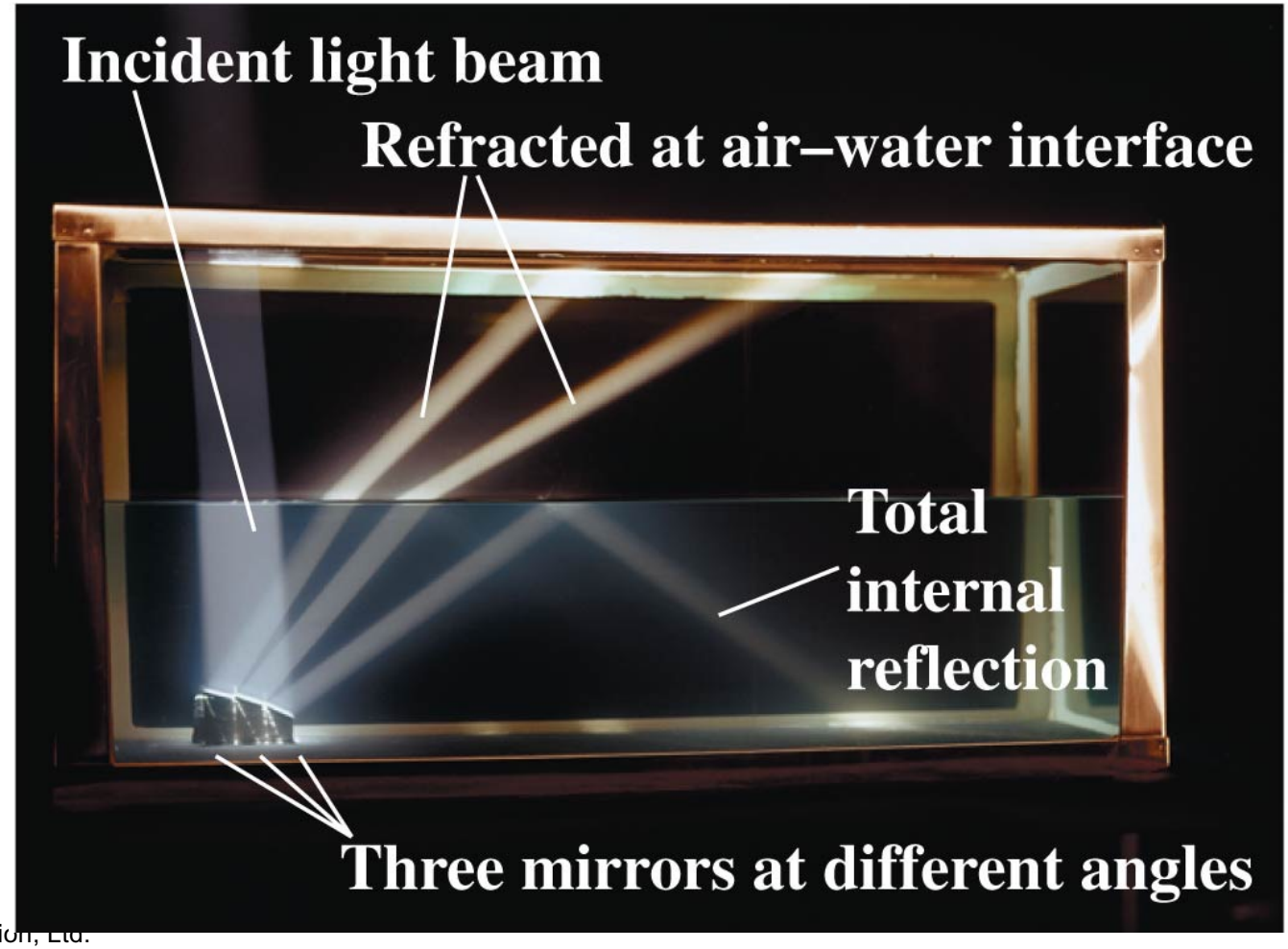




납작한 타원형

© 2016 Pearson Education, Ltd.

# Total internal reflection

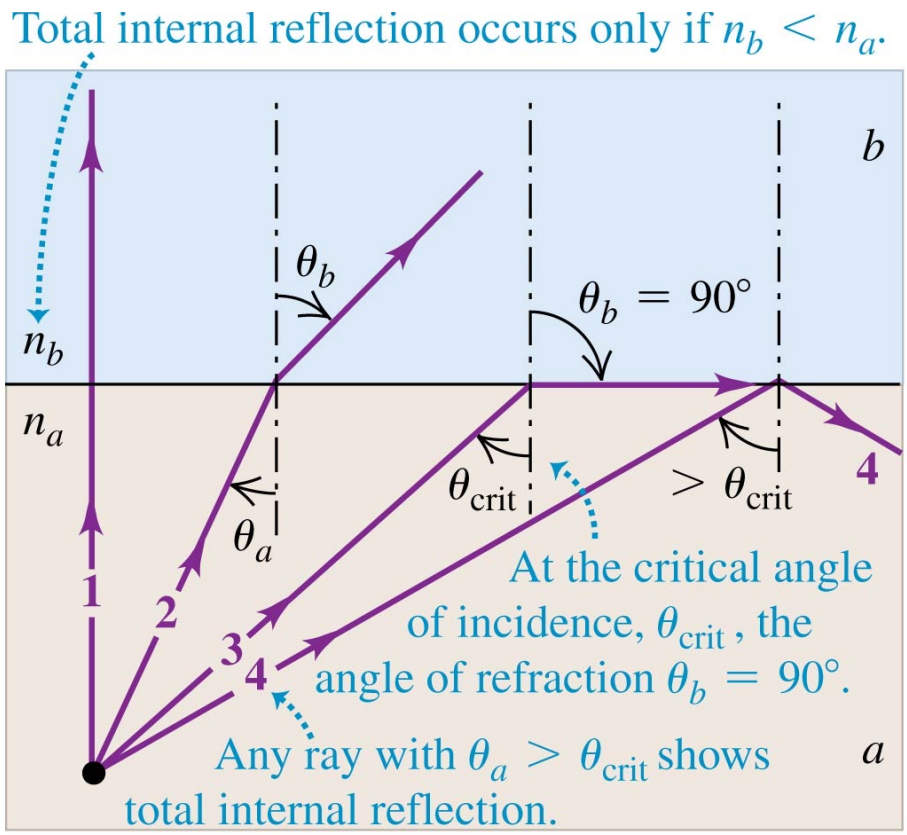
- Under certain circumstances, *all* of the light can be reflected back from an interface, even though the second material is transparent.




© 2016 Pearson Education, Ltd.

# Total internal reflection

- If the angle of incidence is larger than a **critical angle**, the ray cannot pass into the upper material; it is completely reflected at the boundary surface.
- This situation occurs only when $n_b < n_a$.




**Critical angle for total internal reflection**

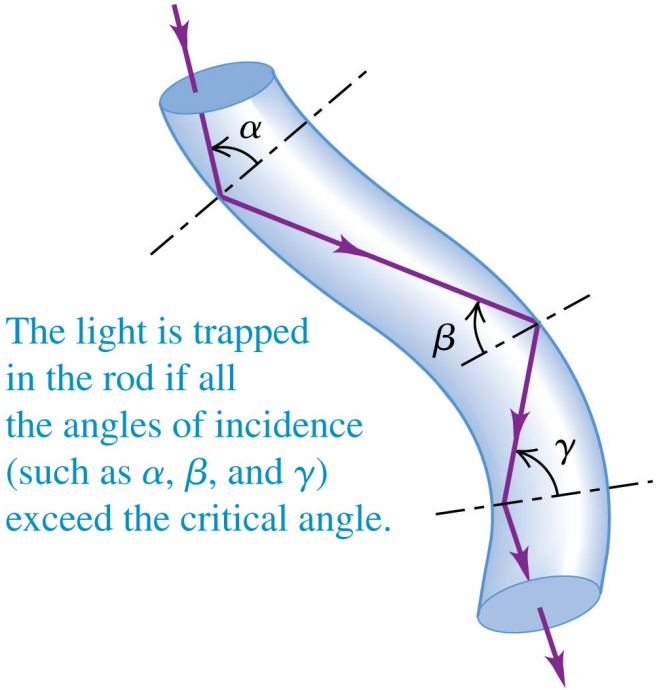
$$\sin\theta_{crit} = \frac{n_b}{n_a}$$

$n_b$: Index of refraction of second material

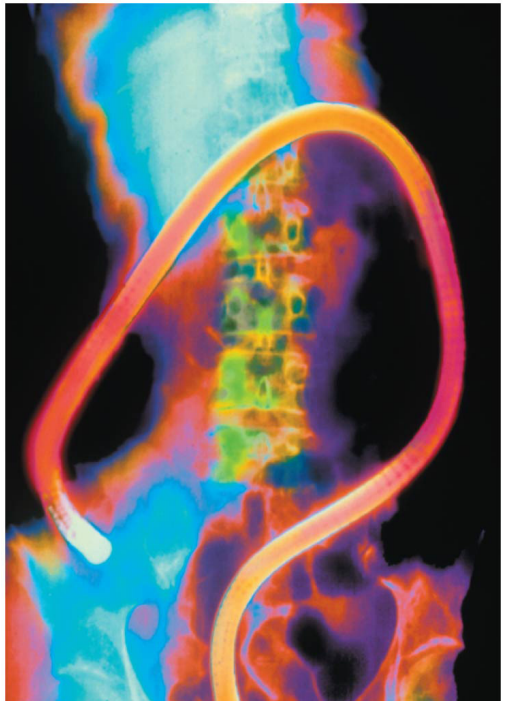
$n_a$: Index of refraction of first material

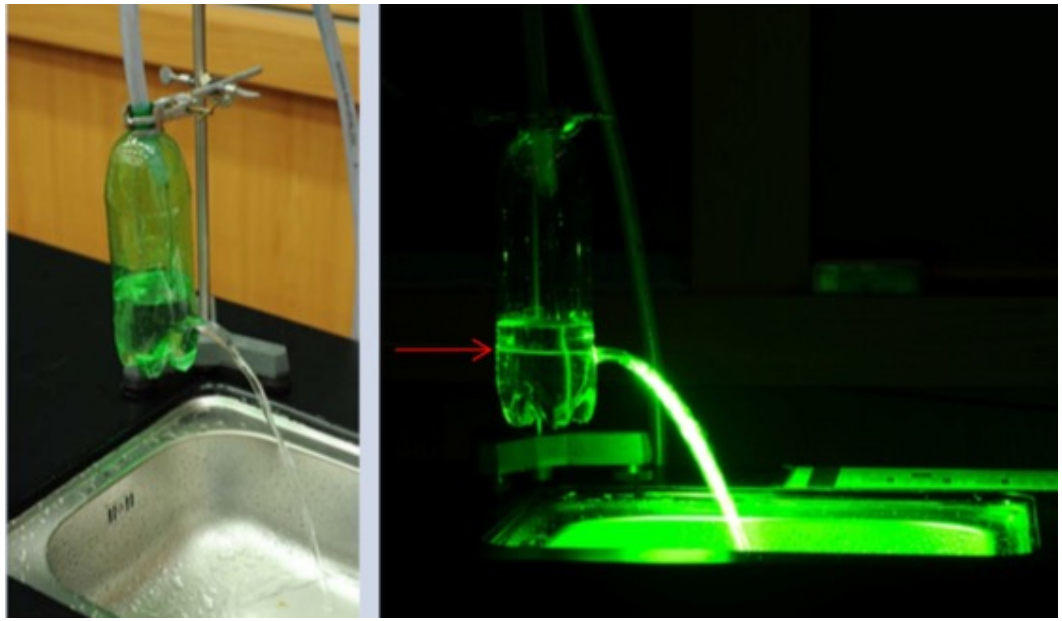
© 2016 Pearson Education, Ltd.

# Fiber optics




The light is trapped in the rod if all the angles of incidence (such as $\alpha$, $\beta$, and $\gamma$) exceed the critical angle.





# Index of refraction and the wave aspects of light

- The frequency $f$ of a wave does not change when passing from one material to another.
- In any material, $v = \lambda f$; since $f$ is the same in any material as in vacuum and $v$ is always less than the wave speed $c$ in vacuum, $\lambda$ is also correspondingly reduced.

**Wavelength of light in a material** $\lambda = \dfrac{\lambda_0}{n}$

$\lambda_0$: Wavelength of light in vacuum

$n$: Index of refraction of the material

- When a wave passes from one material into a second material the waves get "squeezed" (the wavelength gets shorter) if the wave speed decreases and get "stretched" (the wavelength gets longer) if the wave speed increases.

© 2016 Pearson Education, Ltd.